Software Requirements Specification for FPVR: subtitle describing software

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

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(G) Goals

Provided description: Goals are "needs of the target organization, which the system will address". While the development team is the principal user of the other books, the Goals book addresses a wider audience: essentially, all stakeholders [1]

Provided description: It must contain enough information to provide — if read just by itself — a general sketch of the entire project. To this effect, chapter G.3 presents a short overview of the system and G.1 will typically include some key properties of the environment. As it addresses a wide readership, it should be clear and minimize the use of specialized technical terms. Together, G.1, G.2 and G.3 describe the rationale for the project. It is important to state these justifications explicitly. Typically, they are well understood at the start of the project, but management and priorities can change [1]

G.1 Context and Overall Objectives

Provided description: High-level view of the project: organizational context and reason for building a system. It explains why the project is needed, recalls the business context, and presents the general business objectives. [1]

G.2 Current situation

Provided description: Current state of processes to be addressed by the project and the resulting system. It describes the current situation, upon which the system is expected to improve [1]

G.3 Expected Benefits

Provided description: New processes, or improvement to existing processes, made possible by the project's results. It presents the business benefits expected from the successful execution of the project. This chapter is the core of the Goals book, describing what the organization expects from the system. It ensures that the project remains focused: if at some stage it gets pushed in different directions, with "creeping featurism" threatening its integrity, a reminder about the original business goals stated in those chapters will help. [1]

G.4 Functionality overview

Provided description: Overview of the functions (behavior) of the system. Principal properties only (details are in the System book). It is a short overview of the functions of the future system, a kind of capsule version of book S, skipping details but enabling readers to get a quick grasp of what the system will do. [1]

G.5 High-level usage scenarios

Provided description: Fundamental usage paths through the system. It presents the main scenarios (use cases) that the system should cover. The scenarios chosen for appearing here, in the Goals book, should only be the main usage patterns, without details such as special and erroneous cases; they should be stated in user terms only, independently of the system's structure. Detailed usage scenarios, taking into account system details and special cases, will appear in the System book (S.4). [1]

G.6 Limitations and Exclusions

Provided description: Aspects that the system need not address. It states what the system will not do. This chapter addresses a key quality attribute of good requirements: the requirements must be delimited (or "scoped"). G.6 is not, however, the place for an analysis of risks and obstacles, which pertain to the project rather than the goals and correspondingly appears in chapter P.6. [1]

G.7 Stakeholders and requirements sources

Provided description: Groups of people who can affect the project or be affected by it, and other places to consider for information about the project and system. It lists stakeholders and other requirements sources. It should define stakeholders as categories of people, not individuals, even if such individuals are known at the time of writing. The main goal of chapter G.7 is to avoid forgetting any category of people whose input is relevant to the project. It also lists documents and other information that the project, aside from soliciting input from stakeholders, can consult for requirements information. [1]

(E) Environment

Provided description: The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate. [1]

E.1 Glossary

Provided description: Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. It introduces the terminology of the project; not just of the environment in the strict sense, but of all its parts. [1]

E.2 Components

Provided description: List of elements of the environment that may affect or be affected by the system and project. It includes other systems to which the system must be interfaced. These components may include existing systems, particularly software systems, with which the system will interact — by using their APIs (program interfaces), or by providing APIs to them, or both. These are interfaces provided to the system from the outside world. They are distinct from both: interfaces provided by the system to the outside world (S.3); and technology elements that the system's development will require (P.5). [1]

E.3 Constraints

Provided description: Obligations and limits imposed on the project and system by the environment. This chapter defines non-negotiable restrictions coming from the environment (business rules, physical laws, engineering decisions), which the development will have to take into account. [1]

E.4 Assumptions

Provided description: Properties of the environment that may be assumed, with the goal of facilitating the project and simplifying the system. It defines properties that are not imposed by the environment (like those in E.3) but assumed to hold, as an explicit decision meant to facilitate the system's construction. [1]

E.5 Effects

Provided description: Elements and properties of the environment that the system will affect. It defines effects of the system's operations on properties of the environment. Where the previous two categories (E.3, E.4) defined influences of the environment on the system, effects are influences in the reverse direction. [1]

E.6 Invariants

Provided description: Properties of the environment that the system's operation must preserve, i.e., properties of the environment that operations of the system may assume to hold when they start, and must maintain. [1]

(S) System

Provided description: The System book refines the Goals one by focusing on more detailed requirements. It describes the system itself: its components, its functionality, and the criteria by which an implementation will be judged satisfactory. [1]

S.1 Components

Provided description: Overall structure expressed by the list of major software and, if applicable, hardware parts. [1]

S.2 Functionality

Provided description: This is the bulk of the System book, describing elements of functionality (behaviors). This chapter corresponds to the traditional view of requirements as defining "what the system does". It is organized as one section, S.2.n, for each of the components identified in S.1, describing the corresponding behaviors (functional and non-functional properties). [1]

S.3 Interfaces

Provided description: How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs). It specifies how that functionality will be made available to the rest of the world, including people (users) and other systems. These are interfaces provided by the system to the outside; the other way around, interfaces from other systems, which the system may use, are specified in E.2. [1]

S.4 Detailed usage scenarios

Provided description: Examples of interaction between the environment (or human users) and the system, expressed as user stories. Such scenarios are not by themselves a substitute for precise descriptions of functionality (S.3), but provide an important complement by specifying cases that these behavior descriptions must support; they also serve as a basis for developing test cases. The scenarios most relevant for stakeholders are given in chapter G.5 in the Goals book, at a general level, as use cases; in contrast, S.4 can refer to system components and functionality (from other chapters of the System book) as well as special and erroneous cases, and introduce more specific scenarios. [1]

S.5 Prioritization

Provided description: Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality. It is useful in particular if during the course of the project various pressures force the team to drop certain functions. [1]

S.6 Verification and acceptance criteria

Provided description: Specification of the conditions under which an implementation will be deemed satisfactory. Here, "verification" as shorthand for what is more explicitly called "Verification & Validation" (V&V), covering several levels of testing — module testing, integration testing, system testing, user acceptance testing — as well as other techniques such as static analysis and, when applicable, program proving. [1]

(P) Project

Provided description: The Project book describes not the system but the process that produces it: roles, schedule, tasks and deliverables, required technology, and the risks that could derail it. [1]

P.1 Roles and personnel

Provided description: Main responsibilities in the project; required project staff and their needed qualifications. It defines the roles (as a human responsibility) involved in the project. [1]

P.2 Imposed technical choices

Provided description: Any a priori choices binding the project to specific tools, hardware, languages or other technical parameters. Not all technical choices in projects derive from a pure technical analysis; some result from company policies. While some project members may dislike non-strictly-technical decisions, they are a fact of project life and must be documented, in particular for the benefit of one of the quality factors for requirements: "requirements must be justified". [1]

P.3 Schedule and milestones

Provided description: List of tasks to be carried out and their scheduling. It defines the project's key dates. [1]

P.4 Tasks and deliverables

Provided description: This is the core of the Project book. It details the individual tasks listed under P.3 and their expected outcomes. It define the project's main activities and the results they must produce, associated with the milestone dates defined in P.3. [1]

P.5 Required technology elements

Provided description: External systems, hardware and software, expected to be necessary for building the system. It lists external technology elements, such as program libraries and hardware

devices, that the project is expected to require. Although the actual use of such products belongs to design and implementation rather than requirements, it is part of the requirements task to identify elements whose availability is critical to the success of the project — an important element of risk analysis (P.6). [1]

P.6 Risk and mitigation analysis

Provided description: Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise. It is essential to be on the lookout for events that could derail the project, and devise mitigation strategies. It can include a SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) for the project. [1]

P.7 Requirements process and report

Provided description: Initially, description of what the requirements process will be; later, report on its steps. It starts out as a plan for conducting the requirements elicitation process, but is meant to be updated as part of that process so that it includes the key lessons of elicitation. [1]

References

[1] Bertrand Meyer.
Handbook of Requirements and Business Analysis.
Springer, 2022.

[2] Ian Sommerville and Peter Sawyer.
Requirements Engineering: A Good Practice Guide.
Wiley, 1997.

Appendix — Reflection

[Not required for CAS 741 —SS]

The information in this section will be used to evaluate the team members on the

graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?